

Zurich ServiceNow AI Platform Capabilities

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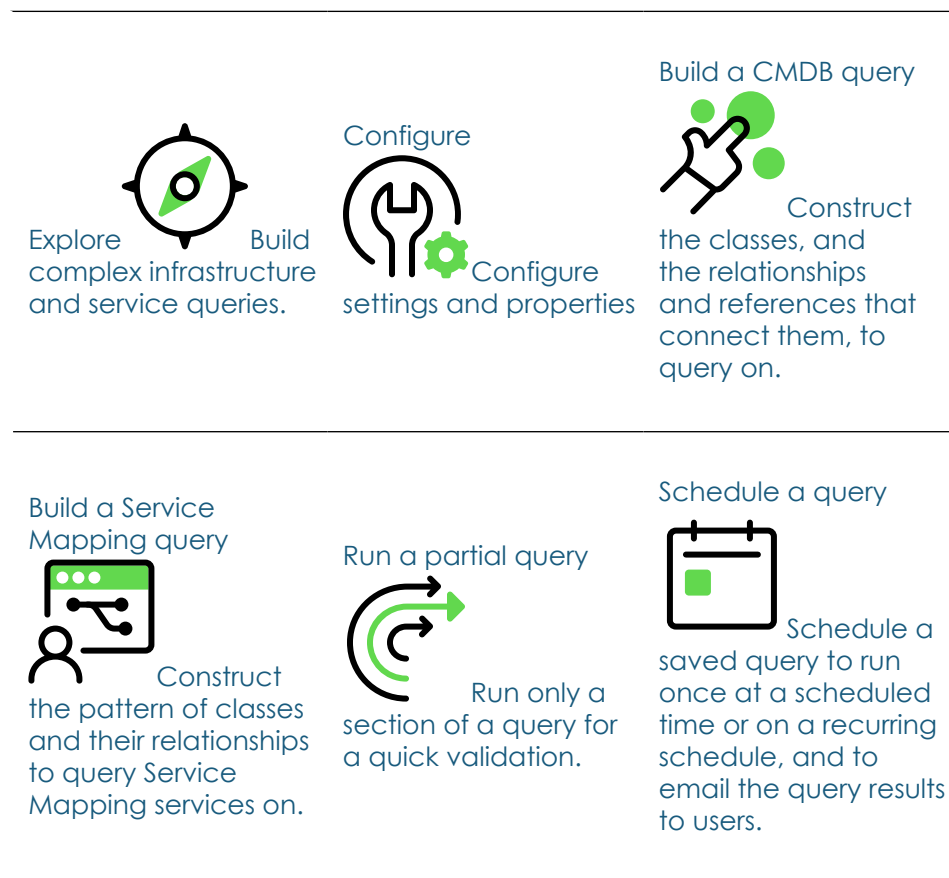
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CMDB Query Builder

The CMDB Query Builder enables you to build complex infrastructure and service queries that span multiple CMDB classes, non-CMDB tables, and that involve many CIs that are connected by different relationships.

Get started

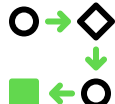


Create a report



Create a basic report, or a dynamic report that continuously updates with the latest query results and add it to Performance Analytics dashboards.

Export or import queries



Export and then import and run a saved query definition between instances, such as between a development and a production environment.

Reference



Reference topics provide example queries and descriptions of navigation and property settings.

Troubleshoot and get help

- For tips and basic troubleshooting, see [CMDB Query Builder \[KB0681251\]](#).
- [KB0546686: CMDB Resources Page](#)
- Search the [Known Error Portal](#) for known error articles
- Contact Customer Service and Support

Exploring CMDB Query Builder

The CMDB Query Builder enables you to build complex infrastructure and service queries that span multiple CMDB classes, non-CMDB tables, and that involve many CIs that are connected by different relationships.

The CMDB Query Builder provides a canvas into which you drag the CI classes that you want to include in a query. Then you add relationships, AND/OR operators between the CI classes, and define the relationship properties to query for. You can use saved queries to populate a CMDB group with CIs, and then use scriptable APIs to retrieve the CI list and apply actions collectively to all the CIs in the group.

There are two query types: CMDB Query and a Service Mapping query, which you can use separately or in combination to create queries such as:

- All hardware in my service offering that has Windows installed.
- All CIs of a certain type in an application service. For example, all Apaches/Web Servers/Linux servers per service.
- All virtual servers and the physical servers that host them.
- All servers that are not mapped to any application service.
- All application services and their associated servers and the cost of each server. This query helps evaluate the cost of technology for each application service.

Starting node: The starting point of the query which is labeled as **STARTING NODE** on the Query Builder canvas. The first class that you drag to the canvas becomes automatically the starting node of the query and you cannot select a different starting node. In a complex query, the starting node must always be the only node connected to an AND/OR operator. If you try to connect a second node to an operator that the starting node is connected to, the query fails to run and a prompt to select a different starting node appears.

Additional information

For a webinar, see [CMDB Query Builder Queries and Reporting - Platform Analytics Academy](#) blog post in the ServiceNow Community.

Intelligent Search for CMDB integration

By default, [Intelligent Search for CMDB](#) functionality is integrated into the CMDB Query Builder. When opening the Query Builder, you can use the Intelligent Search search box which appears above the Query Builder canvas. Intelligent Search lets you use everyday natural language query (NLQ) to build a query. Intelligent Search parses, resolves any ambiguities in table names and relationship types, and then converts your search string into a valid query. The query appears fully constructed on the Query Builder canvas where you can run or continue and develop the query.

The integration of Intelligent Search for CMDB with the CMDB Query Builder is controlled by the system property `glide.cmdb.query.nlq.activated`, which is set to **true** by default. If you set the property to **false**, Intelligent Search for CMDB will not be available within the Query Builder.

AI Search and CMDB

NLQ Genius Results for [AI Search](#) use natural language queries to find CMDB tables that match your search without leaving a portal or workspace. For more information, see [NLQ Genius Results](#).

Note: AI Search is a complementary search experience that allows you to surface CMDB records in your normal flow of work, without having to access the CMDB Query Builder. It doesn't replace intelligent Search for CMDB, which is purpose-built and integrated into CMDB Query Builder.

CMDB Query

A query type that queries the infrastructure for CI classes and the relationships and references that connect them. You can optionally add the context of non-CMDB tables to a CMDB query.

You can include [Application services](#) in a CMDB query, to find, for example:

- All critical application services in your database.
- All infrastructure in a particular application service.
- All incidents for a particular CI in an application service, or all incidents for all the CIs of an application service.
- All application services with a pattern of a service connected to a database, and where the database has incidents.

The list of available non-CMDB tables includes a subset of tables within the system, which have a reference to the Configuration Item [cmdb_ci] class or its children. The list of non-CMDB tables, includes tables such as Asset, Task, and Problem. You can use the system property [glide.cmdb.query.non_cmdb.black_listed_tables](#) to narrow down the list of non-CMDB tables to choose from.

Service Mapping Query

A query type that queries application services. The query is framed within an application service map. You define a pattern, and query for application service maps that have that pattern in their definition. The relationships in Service Mapping queries are matched by single-level direct relationships which is similar to the CMDB queries, and in addition,

they are also matched by multi-level indirect relationships if they exist. A query for a relationship between two CI classes is satisfied even if the two CI classes are connected by intermediate CI classes that are not specified in the query.

Combination Query

You can combine the two query types by incorporating a saved Service Mapping query into a CMDB query. For example, create a CMDB query for Windows Servers that are connected to Tomcat WAR. Then connect the Tomcat WAR CI class to a Service Mapping query. The query changes to find Windows Servers that are connected to Tomcat WAR which is included in the services that returned by the Service Mapping query. You can inverse that query by choosing **Does Not Belong To Service**. This changes the query to find Windows Servers that are connected to Tomcat WAR that is not included in services returned by the Service Mapping query.

Related concepts

- [CMDB groups](#)

Configuring CMDB Query Builder

You can configure settings that affect some functionality of the CMDB Query Builder.

Configuration overview

Configure CMDB Query Builder settings

Configure settings that control the appearance and operation of the CMDB Query Builder canvas.

Configure the relationships to query on

Configure relationships between CIs, such as the parent/child (direction) relationship or first- or second-level relationships.

Configure batch size for queries

If necessary, override the default global batch size (100) that is allocated for every Query Builder query run. Or, optimize the batch size value per saved query.

Set execution mode for running a query

Set the query execution mode for running a query, to use either the legacy engine or an performance-enhancing engine.

Configure CMDB Query Builder settings

Configure settings that control some of the functionality of the CMDB Query Builder.

Before you begin

Role required: admin

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. On the **CMDB Query Builder** page, select the Settings icon () and then configure the settings.

Setting	Description
Display Relationships in Results	Display the relationship between CIs in the query results.
Display Suggested Connections	This setting applies only to CMDB queries. Filter the CMDB classes and the non-CMDB tables lists in the side-panel to display only classes and tables with which the selected

Setting	Description
	node on the canvas has a relationship. You can then drag any item from the filtered list to the canvas and connect it to the selected node on the canvas.
Display Results in New Tab	Display query results in a separate browser tab titled Query Results.

3. Select the Settings icon again to close the dialog box.

Configure the relationships to query on

Configure the type of relationships that you want to query for in CMDB Query Builder. For example, configure which CI on the canvas is the parent and which is the child in the relationship or the level of relationships to query on.

Before you begin

Role required: cmdb_query_builder_read

About this task

When you connect CI classes on the canvas, the CMDB Query Builder displays the Connection Properties in the right-side bar, where you can configure the properties of the relationship to query on. For Service Mapping queries, you can also configure whether to query for related or unrelated CIs.

Note: Newly added relationships between CI classes might take up to 30 minutes to appear in the relationship list.

Procedure

1. While adding CIs to the canvas, select a connection box on a CI and then select the connection box on another CI that you want to create a relationship with.
2. Configure the relationship properties in the Connection Properties side-bar.

Section	Field	Description
Relationship Direction	Parent Child	Set the relationship direction. Which CI class is the parent and which CI class is the child in the relationship.
Relationship Levels	Level	Query only on first-level relationship or also on second-level relationships.
Relationship Types and Related Items	No Relationships	Select to query for CIs which have no relationships to the set class.
Relationship Types and Related Items	Select Add Relationship Types	Query for CIs and descending classes with specific relationship types.
Relationship Types and Related Items	Select Add a Related Item	Query on related CIs using a field that the parent and ancestor parent CI classes use to reference the child CI class.

3. Select **Save**.

Related concepts

- [CMDB groups](#)

Batch size for CMDB Query Builder queries

In a base system, a global batch size of 100 is allocated for every Query Builder query run. If needed, you can use a system property to override the default global batch size, or optimize the batch size value per saved query.

Queries can differ widely as they can be configured to query a wide variety of classes. Therefore, the batch size in the base system might not be optimal for every query, and some queries might time out or take a long time to complete. The optimal batch size for running queries depends on system load such as amount of data and number of relationships in your system. Contact Support for assistance in calculating the batch size for your query.

Batch size is applicable and behaves the same in all query run scenarios, regardless of how the run was initiated:

- Query Builder user interface (ad hoc or saved query)
- [Query Builder Scriptable API](#)
- Scheduled jobs
- [CMDB groups](#)

The batch size for query runs is allocated in the following priority order:

1. The value in the Execution Batch Size field in the Saved Queries table, for a specific saved query. If set, this value applies only to the saved query, and has priority over the global value of 100 and the value of the `glide.cmdb.query.batch_size` system property.
2. The value of the system property `glide.cmdb.query.batch_size`, if exists, determines globally the batch size that is allocated to all query runs. If you add and set this property, the value applies to all queries, other than saved queries with Execution Batch Size value set.
3. A global value of 100, if the previous two options are not configured.

Modifying batch size for queries

If you are experiencing performance problems when running queries, you can modify the batch size value:

- Globally for all queries: By adding (if necessary) and setting the value of the system property [glide.cmdb.query.batch_size](#).
- Per saved query: [Set batch size for a specific saved query](#)

Set batch size for a specific saved query

Configure a custom batch size for a CMDB Query Builder saved query that takes a long time to complete or that times out. A custom batch size overrides the global batch size in the base system and the value of the [glide.cmdb.query.batch.size](#) system property.

Before you begin

Role required: admin

About this task

Contact Support for assistance in calculating the batch size for your query.

Procedure

1. In the Filter navigator, enter `qb_saved_query.list` and press Enter to navigate to the Saved Queries table.
2. In the Saved Queries list view, locate the saved query for which you want to change batch size.
3. Set or modify the value in the Execution Batch Size field. Set the value to be greater than the global value in the [glide.cmdb.query.batch.size](#) property, or increase any existing value.

Set execution mode for running a query

Configure the execution mode to use when running a saved query. The execution mode setting determines whether the query runs using a performance-enhancing engine or the legacy engine.

Before you begin

Role required: cmdb_query_builder

About this task

A saved query can run in either of the following execution modes:

- v1: Runs the query using the legacy query execution engine.
- v2: Runs the query using an enhanced query execution engine, which is designed for improved performance and scalability.

Note: When running queries that contain OR nodes, the format of the returned results is different between the options.

The system uses query-level settings and system properties to determine which mode to use, as follows:

1. Check the query setting in the Query Execution Mode attribute in the Saved Queries [qb_saved_query] table:
 - If '**v1**' or 'v2': Use the specified value.
 - If '**None**': Check system properties as described in the next step.
2. Check system property glide.cmdb.query.execution_mode:
 - If '**v1**', null, or empty: Use v1
 -

If '**v2**': Perform 'OR'-specific evaluation:

If the query contains one or more logical OR conditions:

- Check the value of glide.cmdb.query.or_execution_mode:
 - If '**v1**' or '**v2**': Use that value
 - If empty: Use v1
- If the query contains no OR conditions: Use v2

For more information about the `glide.cmdb.query.execution_mode` and the `glide.cmdb.query.or_execution_mode` system properties, see [Property settings for CMDB Query Builder](#).

Procedure

1. Navigate to **All** and then, in the navigation filter, enter `qb_saved_query.list` to open the Saved Queries table.
2. In the Saved Queries list view, select the query for which you want to set execution mode.
3. On the query's form, set **Query Execution Mode** to 'v1' or 'v2'.
4. Select **Update**.

Navigation in CMDB Query Builder

Use the navigation tools to enlarge or shrink the query, to move the query, or to border a section of the query to run.



Use the buttons in the navigation tool as follows:

- Use the plus sign (+) to increase magnification of the query.
- Use the minus sign (-) to decrease magnification of the query.
- Select the center dot to center the query on the canvas.
- Use the direction arrows to move the query in that direction.
- Use the selection tool under the navigation tool to toggle between two states:
 - Moving the entire query on the canvas.

- Bordering a section of the query, which you can then run as a partial query.

Build a CMDB query using the CMDB Query Builder

A CMDB query type that queries the infrastructure for CI classes and optionally non-CMDB tables, and the relationships and references that connect them.

Before you begin

Role required: `cmdb_query_builder_read` to only view and run saved queries, and `cmdb_query_builder` (contained for `itil`, `itil_admin`, and `asset`) to create and save queries, modify saved queries, and run queries. For reporting on queries, report related ACLs are required.

Authorized users can update and [delete](#) a query created by another user.

About this task

Build the query by dragging the CI classes and non-CMDB tables that you want to include in the query. Then dropping them as nodes on the canvas, and defining relationship properties between them. You can filter on the attributes of any node to narrow down the results to a specific set of CIs of that class or to a single specific CI. You can also select which property columns appear in the query results.

As you step through building a query, list options and other user interface elements of the CMDB Query Builder, are dynamically filtered as appropriate to your selections.

To learn more about using Query Builder, see the [CMDB Query Builder Queries and Reporting - Platform Analytics Academy](#) ServiceNow Community video. For a step-by-step walk through of building CMDB queries in the Query Builder, including queries with application services, see [Example queries](#).

Procedure

1. Navigate to **All > Configuration** and select **CMDB Query Builder**.
2. On the **CMDB Query Builder** page do either of the following steps:

- Select **Create new**. Type in a **Name**, choose **CMDB Query** as the **Query type**, and then select **Create**.
 - Select a widget of a saved query to continue building an existing query. [Search for a saved query](#) first if needed.
 - Point to the upper right corner of a saved query widget, and select the **Duplicate Query** icon to edit a copy of a saved query. The default name of the new query contains the string 'copy'.
3. On the canvas, you can do any of the following operations:
- Add CI classes to the query: On the **CMDB Classes** tab, select classes from the hierarchy list and drag them to the canvas.
 -
- Add an Application Service CI to the query: On the **CMDB Classes** tab, select the **Application Service** class from the hierarchy list and drag it to the canvas.
- In Application Service Properties on the right-side bar, you can select **Convert attached nodes to pattern** to query on patterns between the application service and other CMDB class nodes. When querying on a pattern, the nodes on both ends of the pattern connection, can be any number of levels apart. If **Convert attached nodes to pattern** is not selected, then the connection between the application service node and other CMDB class nodes, represent direct relationships.
- Add non-CMDB tables to the query: Select a table from the **Non-CMDB Tables** list and drag it to the canvas.
- Note:** A non-CMDB table cannot be the starting node in the query.
-
- Add connections (relationships or patterns for application services) between two nodes on the canvas:
- a. On the first node in the connection, select the small square at the center of the right side.

- b. On the second node in the connection, select the small square at the center of the left side to create the connection.

Connection UI Notations

Notation	Description
Full line	A relationship in a CMDB query.
Red asterisk at the center of the connection line	Information such as relationship type is missing, invalidating the query.
Levels:<n> Types:<n> or a <Reference type> notation on the connection line	As applicable: The number of relationship levels and the number of relationship types included for the connection. Or, a reference type for a relationship that is a reference.
Dashed line	A pattern connection between an application service node and another node.

- In Connection Properties on the right-side bar, configure relationship settings (select the connection line if necessary):
 - a. In the Relationship Direction section, select the **Parent** node (the **Child** node automatically adjusts).
 - b. In the Relationship Levels section, set **Level** to **First level relationships** if the CIs are directly connected. Or, **Up to 2nd level relationships** if the CIs are connected either directly or indirectly through another CI.
 - c. In the Relationship Types and Related Items section, select either option:

Option	Description
No Relationships	To query for CIs with no connecting relationships, such as All Tomcat WAR CIs which are not connected to a Windows Server.
Add Relationship Types	To select specific or any relationship type.
Add a Related Item	To query for related CIs between the nodes.

- Configure CI reference column for a connection to a non-CMDB table: In the Connection Properties right-side bar, in the CI Reference Column section, select the column with a reference to a CI from the **Use CI reference column** list. If only one option is available, it is automatically selected.
- Configure the pattern between an application service node set with the **Convert attached nodes to pattern** option, and a non-CMDB table node: Select **Apply <table> reference filter to all nodes in the pattern** to apply the query to the application service CI itself and to the CIs within the application service.
- Add filters to a class node: Apply filters to narrow down a class query to a specific set of CIs or to a single specific CI.
 - a: Point to the node to add a filter to, and then select the **Apply filters** icon that pops up above the node.
 - b: In the Filters section, add attribute and [related list conditions](#).
 - c: Close the **Filters** section.

For example: Add a filter for database location to query for databases located in Seattle.

Select **Applied Filters** in the right-side bar to view all filters for each node on the canvas.

- Add And/Or operators to the query:

- a: Connect one node to two other nodes.
 - b: Select the **And** box that appears on the connection line, to toggle between the **And/Or** operators.
For example: C1 is Tomcat WAR, C2 is Linux Server, and C3 is Windows Server. Query for All Tomcat WAR CIs which are connected either to Linux Server Or to a Windows Server.
- Add property columns for a node, to appear in the query results:
 - Note:** For a relationship, the query results include the parent, child, and type columns. You cannot add any other columns from the [cmdb_rel_ci] table.
- a: Select **Properties** in the right-side pane.
 - b: Select a node once or twice, so that the Report Columns section appears in the right-side bar, and then select **Add Columns**.
 - c: Select properties and then select outside the properties list to close it.
- Create a combination query by integrating a Service Mapping query into a CMDB query:
 - a: When building a CMDB query, select **Saved Service Queries** in the left-side bar.
 - b: Select and then drag a Service Mapping query to the canvas.
This query returns all CIs that satisfy the CMDB query, and that are included in the services returned by the embedded Service Mapping query.
- Add a search tag that can then be used as a search criteria for saved queries:
 - a: Select the **Add Tags** icon at the top of the canvas.
 - b: Select **Add Tag** and in the **Query Tags** dialog box enter one or more tag strings.

- c. Select the **Add Tags** icon again to close the **Query Tags** dialog box.

4. Select **Save**.

On the **Saved Queries** tab, point to a saved query widget and select **Query Information**. Query details such as the query type, last update date, CMDB groups associated with the query, and the query schedules appear.

What to do next



Select **Run**.

Only the first 100 results of the query appear in the results pane.

- Select **Load More Results** to view the next set of 100 results.
- Select **Load All Results** to view the rest of the query results, up to the number specified by the [glide.cmdb.query.max_results_limit](#) system property (10,000 by default).

Select a CI to open its CI form, and on the CI form select **Open in CMDB Workspace** to view CI health in the CI Health tile in CMDB Workspace. This operation requires the `sn_cmdb_user` (CMDB user) role.

Note:

- Ensure that the [glide.security.use_csrf_token](#) property is set to true, allowing all results to appear.
- When **Level** is set to **Up to 2nd level relationships**, the relationship type does not appear in the query results.
- When a query is running, wait for it to complete or to time out before opening or running another query.

- **Modify Query Builder settings:** Select the  (Settings) icon to open the **Query Builder Settings** dialog box.

- Copy and share the URL of a saved query with users that have access to the CMDB Query Builder. Pasting the shared URL in a new internet browser window, directly opens the saved query in the CMDB Query Builder.
- [Create a report in CMDB Query Builder.](#)
- [Create a schedule](#) to run the query at a future time, and to email the results to interested parties.
- Export query results:

Select the Query Results context menu and select **Export**. Even if the **Load More Results** button is visible, indicating that there are additional query results, only the results that are visible are exported.

- [Export and import a CMDB query](#) to port a query definition between instances.
- [Populate a CMDB group](#) using a saved query.
- [Delete a CMDB query.](#)

Related concepts

- [Example queries](#)

Build a Service Mapping query using the CMDB Query Builder

The Service Mapping query type is a pattern consisting of classes and relationships between those classes. After you build the pattern and run the query, the query returns all the Service Mapping services that contain that pattern.

Before you begin

Role required: `cmdb_query_builder_read` to only view and run saved queries, and `cmdb_query_builder` (contained for `itil`, `itil_admin`, and `asset`) to create and save queries, modify saved queries, and run queries.

Authorized users can update and [delete](#) a query created by another user.

About this task

Build the query by dragging the CI classes that you want to include in the query, dropping them as nodes on the canvas, and then defining relationship properties between them. For every class node in the query, you can filter on its attributes to narrow down the results to a specific set of CIs of that class or to a single specific CI. You can also select which property columns appear in the query results.

As you step through building a query, list options and other user interface elements of the CMDB Query Builder, are dynamically filtered as appropriate to your selections.

See [Example queries](#) for a step-by-step walk through of building a Service Mapping query in the CMDB Query Builder.

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. On the **CMDB Query Builder** page do either of the following:
 - a. Select **Create new**.
 - b. Enter a **Name**.
 - c. Select **Service Mapping Query** as the **Query type**.
 - d. Select **Create**.
 - e. Select a widget of a saved query to continue building an existing query. [Search for a saved query](#) first if needed.
 - f. Point to the upper right corner of a saved query widget.
 - g. Select the **Duplicate Query** icon to edit a copy of a saved query. The new query's default name contains the string 'copy'.
3. On the canvas, you can do any of the following:
 - Add CI classes to the query: Select classes from the **CMDB Classes** hierarchy list and drag them to the canvas.
 - Add connections (relationships) between two nodes on the canvas:

- a: On the first node in the relationship, select the small square at the center of the right side.
- b: On the second node in the relationship, select the small square at the center of the left side to create the connection.
- c: In **Connection Properties** on the right-side bar, configure the following (select the connection line if necessary):
 - In the Relationship Direction section, select the **Parent** node (the **Child** node automatically adjusts).
 - In the Service Query Properties section, select **Find Related CIs** or **Find Unrelated CIs** to query for a pattern in which the two classes have or do not have relationships with each other, respectively.

For example, All Tomcat WAR CIs which are not connected to a Windows Server.

Relationship UI Notations

Notation	Description
Dashed line	A relationship in a Service Mapping query.

- Add filters to a class node: Apply filters to narrow down a class to a specific set of CIs or to a single specific CI.
 - a: Point to the node to add a filter to, and then select the **Apply filters** icon that pops up above the node.
 - b: In the Filters section, add attribute and [related list conditions](#).
 - c: Close the **Filters** section.

For example: Add a filter for business criticality to query for businesses that are 'most critical'.

Select **Applied Filters** in the right-side bar to view all filters for each node on the canvas.

- Add And/Or operators to the query:

- a: Connect one node to two other nodes.
- b: Select the **And** box that appears on the connection line, to toggle between the **And** and the **Or** operators.

For example C1 is Tomcat WAR, C2 is Linux Server, and C3 is Windows Server. Query for all Tomcat WAR CIs which are connected either to Linux Server Or to a Windows Server.

- Add property columns for a node, to display in the query results:
 - Note:** For a relationship, the query results display the parent, child, and type columns. You cannot add any other columns from the [cmdb_rel_ci] table.
- a: Select **Properties** in the right-side pane.
- b: Select a node once or twice, so that the node's Report Columns section appears in the right-side bar, and then select **Add Columns**.
- c: Select properties and then select outside the properties list to close it.
- Select columns and add filters that will be applied to the resulting set of services:
 - a: Select **Properties** in the right-side bar, and then select an empty space on the canvas to ensure that nothing is selected.
 - b: Select **Add Columns** at the bottom of the right-side bar and select columns to add. Select outside the columns list to close it.
 - c: Select the **Apply Service Mapping Query Filters** icon at the top of the canvas and add filters.
- Inverse the entire query and search for all Service Mapping services that do not include the query pattern: Select a node once or twice so that Query Properties appear in the right-side bar. In the Metadata section, toggle **Services Including This Pattern** to enable or disable the option.

- Add a search tag that can then be used as a search criteria for saved queries:
 - a: Select the **Add Tags** icon at the top of the canvas.
 - b: Select **Add Tag** and in the **Query Tags** dialog box enter one or more tag strings.
 - c: Select the **Add Tags** icon again to close the **Query Tags** dialog box.

4. Select **Save**.

On the **Saved Queries** tab, point to a saved query widget and select **Query Information** to view query details such as the query type, last update date, and the query schedules.

What to do next

- Select **Run**.
- The query results pane displays only the first 100 results of the query.
- Select **Load More Results** to display the next set of 100 results.
 - Select **Load All Results** to display the rest of the query results, up to the number specified by the `glide.cmdb.query.max_results_limit` system property (10,000 by default).

Select a CI to open its CI form, and on the CI form select **Open in CMDB Workspace** to view CI health in the CI Health tile in CMDB Workspace. This operation requires the `sn_cmdb_user` (CMDB user) role.

Note: When a query is running, wait for it to complete or to time out before opening or running another query.

- **Modify Query Builder settings:** Select the Settings icon () icon to open the **Query Builder Settings** dialog box.

- Copy and share the URL of a saved query with users that have access to the CMDB Query Builder. Pasting the shared URL in a new internet browser window, directly opens the saved query in the CMDB Query Builder.
- [Create a report in CMDB Query Builder.](#)
- [Create a schedule](#) to run the query at a future time, and to email the results to interested parties.
- Export query results:

Export query results that are visible: Select the Query Results context menu and select **Export**. Even if the **Load More Results** button is visible, indicating that there are additional query results, only the results that are visible are exported.

- [Export and import a CMDB query](#) to port a query definition between instances.
- [Populate a CMDB group](#) using the saved query.
- [Delete a CMDB query.](#)

Related concepts

- [Example queries](#)

Run a partial CMDB query

You can run a partial query in the CMDB Query Builder by defining a section of a query, and then running it.

Before you begin

Role required: `cmdb_query_builder` (contained for `itil`, `itil_admin`, and `asset`)

About this task

While building a query or reviewing a saved query, you can run only a section of the query. On the canvas in the CMDB Query Builder, highlight a section of the query which contains the nodes and relationships of the

partial query that you want to run. You can then examine the results of the partial query, and update the query if needed.

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. On the CMDB Query Builder page, select a tile to open an existing query.
3. Select the selection tool under the [navigation tool](#) to switch to a section selection mode.
4. Border a section of the query:
 - a. Select the upper left corner of the section.
 - b. Drag and release at the bottom right corner of the section that you want to create.
As the cursor moves, the selected section is highlight in light blue.
The query nodes that are included in the partial query, appear with a blue border.
5. Select **Run**.
6. In the Pick Starting Node dialog box, select the starting node for the partial query and then select **Confirm**.

Result

The results of the partial query appear in the Results pane.

Delete a CMDB query

Delete a CMDB query that is no longer used or needed.

Before you begin

The query that you want to delete must be already saved.

Role required: cmdb_query_builder (contained for itil, itil_admin, and asset). Authorized users can delete a query created by another user.

Procedure

1. Navigate to **All > Configuration** and select **CMDB Query Builder**.
2. On the **CMDB Query Builder** page, select the Saved Queries tab and set the viewing mode to Card view ().
3. Move to the card with the query that you want to delete.
4. Select the Delete Query (X) icon that appears in the card.

Create a report in CMDB Query Builder

Use CMDB Query Builder reports to show the results of a CMDB query or a Service Mapping query. Create a basic report or a dynamic report that auto-updates when the results of the associated saved query change.

Create a basic report

After running a query in the CMDB Query Builder, you can create a basic report that is scoped to the query execution.

Before you begin

The [Core UI plugin](#) (com.glide.ui.ui16) must be activated.

Role required:

- View report: cmdb_query_builder or cmdb_query_builder_read
- Create report: cmdb_query_builder or cmdb_query_builder_read, and report_user

In a base system, the cmdb_query_builder role is contained in the itil and asset roles.

Procedure

1. Navigate to **All > Configuration** and click **CMDB Query Builder**.
2. Build a query.

3. In the query results pane, click **Load More Results** or **Load All Results** to load all the results that you want to include in the basic report.
4. In the query results pane, click the column context menu and select **Bar Chart** or **Pie Chart**.

Result

The Reports application creates a basic report, which is scoped to the query results that are currently loaded and is static.

Related tasks

- [Build a CMDB query using the CMDB Query Builder](#)
- [Build a Service Mapping query using the CMDB Query Builder](#)
- [Run a partial CMDB query](#)

Related topics

- [Reporting](#)

Create a dynamic report

After running a saved query in the CMDB Query Builder, create a dynamic report that continuously updates to show the latest query results. You can use a dynamic report as any other report created using Reporting and you can add it to Performance Analytics dashboards.

Before you begin

Ensure that the query that you want to create a dynamic report for is a saved query, and that there is a specified schedule for the query. Also, run the saved query and ensure that all query results are visible.

Role required:

- To create: `cmdb_query_builder` and `report_user`
- To view: Reporting role requirements might apply, see [Administering reports](#) for Reporting role requirements.

In the base system, the cmdb_query_builder role is contained in the itil and asset roles.

About this task

The **Create Report** button in CMDB Query Builder which is used to create a dynamic report, is activated only if:

- The query is saved
- The query has a schedule
- The entire set of query results is present after a query run

The initial dynamic report that you create, is based on the results from the initial run of the saved query. Then, on every subsequent run of the saved query, the associated report automatically updates with the latest query results.

However, if you change the query definition itself, the query and the report are no longer in sync and you must create a new report.

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. In the **Saved Queries** tab, select the saved query for which you want to create a report.
Ensure that the query has a specified schedule.
3. Select **Run** and ensure that all query results appear.
Click **Load All Results** if available, to load all results. The **Create Report** button is enabled only if all query results are showing.
4. Click **Create Report**.
If the CMDB Query Builder displays the query results in a new tab, then after the new tab opens with the query results, return to the CMDB Query Builder window.
5. In the Report Designer, click **Next** or **Back** to view and configure the new report in the **Data**, **Type**, **Configure**, and **Style** tabs.
The report is pre-populated with the CMDB query results and a few other report details.
 - **Report name** is set to the name of the saved query.

- **Source type** is set to **Data source**.
- **Data source** is set to the table in which the query results are stored.
- **Query Sys ID** is the ID of the latest run of the query.

For more details about Reporting and about configuring a report in the Report Designer, see [Reporting](#), [Creating reports](#).

6. Click **Save** or **Run**.

Result

CMDB Query Builder creates a report source which you can attach to a report and use with dashboards. For more details about report sources, see [Report sources](#).

What to do next

Use either of the following steps to view the new report source. The name of the new report source is set to the name of the CMDB query it was created from, and cannot be changed.

- In Query Builder, click **Saved Queries**. In the Saved Queries window, click the **Query Information** icon in the tile of the saved query. Scroll to the bottom of the information list and then click the link under **Report source**.
- Navigate to **All > Reports > Administration > Report Sources** and locate the new report source.

Search for a saved query

The CMDB Query Builder allows you to search for a specific saved query using any combination of search criteria such as the query's name, type, custom tags, and who created or updated the query.

Before you begin

To locate a saved query using a **Query Tags** search criteria, the query must have a query tag associated with it. For more information see [Build](#)

a CMDB query using the CMDB Query Builder, or Build a Service Mapping query using the CMDB Query Builder.

Role required: cmdb_query_builder (contained for itil, itil_admin, and asset)

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
The Saved Queries tab shows all saved queries in tiles, and also a tile for creating a new query.
2. On the Saved Queries tab, specify a **Sort by** criteria to filter the saved queries that are showing.
3. In the **Search Saved Queries** field, enter a search string that corresponds to the **Sort by** selection and then select an item from the list that gets populated as you type.
The drop-down list includes search strings that can be applied to the sorting criteria such as 'Query name'.
4. Refine the search by entering additional **Search Saved Queries** search strings as needed.

Schedule a CMDB query

Schedule a saved CMDB query to run once at a scheduled time or on a recurring schedule, and to email the query results to specified users.

Before you begin

A saved CMDB query that was built in the CMDB Query Builder must exist.

Role required: cmdb_query_builder (contained for itil and asset)

About this task

The query results are attached to the email as a file in the specified format. The maximum number of result rows that can be attached is determined by the glide.cmdb.query.max_results_limit system property (10,000 by default).

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder** and then select a saved query.
2. Select **Create Schedule** and then fill in the form.

Scheduled Email of Query Builder form

Field	Description
Query	The query to run.
Users	Users who should receive query results email. To receive emails, users must have an Email address defined and have Notifications set to Enable in their user records.
Groups	Groups to email the query results to.
Zip output	Indicates whether the report should be sent as a zip file.
Active	Indicates whether to run the query according to the specified schedule.
Run	Frequency for running the query.
Time	Time of day to run the query.
Conditional	Indicates whether to display the Condition field, which allows you to specify conditions under which the query runs.

Field	Description
Omit if not records	Indicate whether to distribute email if the query returns zero results.
Email addresses	Email addresses of users who should receive the email but who are not in the system.
Subject	Text that appears in the subject line of the distribution email.
Introductory message	Additional message that is delivered with the query results.
Condition	User-created script that checks for certain conditions to be true before running the query. This field is visible only when Conditional is checked.
Type	File format to use for the query results. Note: Configure the form layout to add this field to the form.

Export and import a CMDB query

Export a saved CMDB or Service Mapping query definition to an XML file which you can later import and run in the CMDB Query Builder. This process lets you port a saved query between instances, such as from a development environment to a production environment.

Before you begin

- You must save a query before you can export it.

- Domain in an exported query must be visible in both, source and destination instances.

Role required: cmdb_query_builder (contained in itil, itil_admin, and asset).

About this task

When exporting a combination query, the integrated Service Mapping query definition is included in the exported query.

For backward compatibility, you can alternatively [Export and import a query as an update set](#).

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. Export a saved query:
 - a. In the **Saved Queries** tab, in either list view or card view, select a saved query.
 - b. Select the **Export query** icon at the top of the Query Builder canvas.
 - c. Wait for the **Query Export Complete** message to appear and then select **Download**.
You can now access the query XML file.
3. Import a saved query:
 - a. In the **Saved Queries** tab, select the **Import query** icon at the top of the CMDB Query Builder window.
 - b. In Finder, select the saved query XML file and select **Open**.
The imported query is available in the **Saved Queries** tab of the CMDB Query Builder on the instance.

Export and import a query as an update set

Export a saved query definition to an XML file as an update set, which you can later import.

Before you begin

Role required: To export — `cmdb_query_builder` (contained in `itil` and `asset`). To import — user with permission to import an update set.

Domain in the exported query must be visible in both, source and destination instances.

About this task

Export a query definition as an update set which you can later import and commit. This process lets you port a query between instances, such as from a development environment to a production environment. For more information about exporting and then committing update sets using XML files, see [Save an update set as a local XML file](#).

When exporting a combination query, the integrated Service Mapping query definition is included in the exported update set.

Procedure

1. Export a saved query:
 - a. In the **Filter navigator**, enter `qb_saved_query.list`.
 - b. Select **Enter** to navigate to the Saved Queries table.
 - c. In the Saved Queries list view, select the query that you want to export.
 - d. Select **Actions on selected rows** and then **Export query**.
 - e. Wait for the **Query Exporter** to complete the export.
2. Import the exported saved query:
 - a. Navigate to **System Update Sets > Retrieved Update Sets**.
 - b. On the Retrieved Update Sets form, select **Import Update Sets from XML**.
 - c. On the Import XML page, select **Choose file**.
 - d. Select the exported XML file.

- e. Select **Upload**.
- f. Open the new record that was added to the Retrieved Update Sets list view.
- g. On the Retrieved Update Set form, select **Preview Update Set Batch**.
- h. Close the Batch Update Set Preview dialog box.
- i. Select **Commit Update Set Batch**.
- j. Close the Commit Update Set Batch dialog box.

Result

The imported query is added to the Query Builder saved queries on the instance.

CMDB Query Builder reference

Reference topics provide example queries and descriptions of navigation and property settings.

Reference topics

Domain separation and CMDB Query Builder

Domain separation is supported in the CMDB Query Builder. Domain separation enables you to separate data, processes, and administrative tasks into logical groupings called domains. You can control several aspects of this separation, including which users can see and access data.

Navigation in CMDB Query Builder

Use the navigation tools to enlarge or shrink the query, to move the query, or to border a section of the query to run.

Property settings for CMDB Query Builder

Use the CMDB Query Builder properties to configure query processing.

Example queries

Use the example queries as models for building your custom CMDB queries and Service Mapping queries.

Domain separation and CMDB Query Builder

Domain separation is supported in the CMDB Query Builder. Domain separation enables you to separate data, processes, and administrative tasks into logical groupings called domains. You can control several aspects of this separation, including which users can see and access data.

Overview

How domain separation works in the CMDB Query Builder

With the CMDB Query Builder you can easily build complex infrastructure and service queries that span multiple CMDB classes, and that involve many CIs that are connected by different relationships. Domain separation is set to be on by default.

- **Saved Query**

The user creates a query by dragging a class node from the class hierarchy and dropping it to the canvas and connecting the nodes with the relationships type.

The user can save the created query as an XML file to the database [qb_saved_query] table in the CMDB for future use. The saved query is domain separated.

- **Query results**

With a saved query, the user selects **Run** and the query result is saved and displays in the platform list view.

In the query results, domain separation behaves in the same way as the platform list view for the CI relationship [cmdb_rel_ci] table and CMDB CI [cmdb] table. Consequently, since the CI relationship is not domain separated, all relationships of the query result display, regardless of the domains. Conversely, if the query result is CI only, since the CMDB CI is domain separated, the results display only if visible in the current domain.

Related concepts

- [Domain separation and Configuration Management Database \(CMDB\)](#)

Property settings for CMDB Query Builder

Set property values to configure query processing.

Access and Role required

The admin role is required to view and edit Query Builder properties.

Note: To open the System Properties [sys_properties] table, enter `sys_properties.list` in the navigation filter.

Properties for CMDB Query Builder

Property	Description
Limits the number of results for a scheduled query and in the results section in the Query Builder when you select Load All Results . glide.cmdb.query.max_results_limit	• Type: integer • Default value: 10000 • Location: Configuration > CMDB Properties > Query Builder Properties
Time limit (in seconds) for running one batch to get one batch of query results (100 results). glide.cmdb.query.batch_time_limit_in_sec	• Type: integer • Default value: 300 • Location: Configuration > CMDB Properties > Query Builder Properties
Time limit (in seconds) for running an entire query to get all results.	• Type: integer • Default value: 1800

Property	Description
glide.cmdb.query.query_time_limit_in_sec	Location: Configuration > CMDB Properties > Query Builder Properties
Exclude list of non-CMDB tables. The specified tables will not appear in the CMDB Query Builder when a user creates a query. glide.cmdb.query.non_cmdb.black_listed_tables	- Type: string - Default value: empty - Other values: Comma-separated list of table names (not labels). Can include '*abc' to exclude all tables containing 'abc' in their table name. - Location: Configuration > CMDB Properties > Query Builder Properties
glide.cmdb.query.batch_size	Batch size allocated globally when saved queries run. - Type: integer - Default value: 100 - Location: Add to System Properties [sys_properties] table.
glide.cmdb.query.execution_mode	Default execution engine to use when running a saved query when the query's execution mode isn't set (Query Execution Mode is 'None'). Used in combination with the glide.cmdb.query.or_execution_mode system property.

Property	Description
	- Type: string - Values: - v1: Use the legacy query execution engine. - v2: Use an enhanced query execution engine which is designed for improved performance and scalability. - Default value: v2 - Location: System Properties [sys_properties] table. - Learn more: Set execution mode for running a query
glide.cmdb.query.or_execution_mode	Default execution engine to use, applying only to queries that contain at least one logical OR condition. Used in combination of the glide.cmdb.query.execution_mode system property. - Type: string - Values: - v1: Use the legacy query execution engine. - v2: Use an enhanced query execution engine which is designed for improved performance and scalability. - Default value: v2

Property	Description
	- Location: System Properties [sys_properties] table. - Learn more: Set execution mode for running a query

Example queries

Use the example queries as models for building your custom CMDB queries and Service Mapping queries.

CMDB — Servers connected to a database

Use this example to build a CMDB query to find all servers with a connection to a database.

CMDB — Critical application services

Use this example to build a CMDB query to find all critical application services, and their owner.

CMDB — Application services with incident or change

Use this example to build a CMDB query to find all application services for which there is an incident or a change request. Results are returned for either the application service itself or any CI within the service.

CMDB — Hardware with Windows installed

Use this example to build a CMDB query to find all hardware in my service offering that has Microsoft Windows installed.

Service Mapping — Linux servers in services

Use this example to build a Service Mapping query to find all Linux servers in services.

CMDB — Servers connected to a database

Use this example to build a CMDB query to find all servers with a connection to a database.

Before you begin

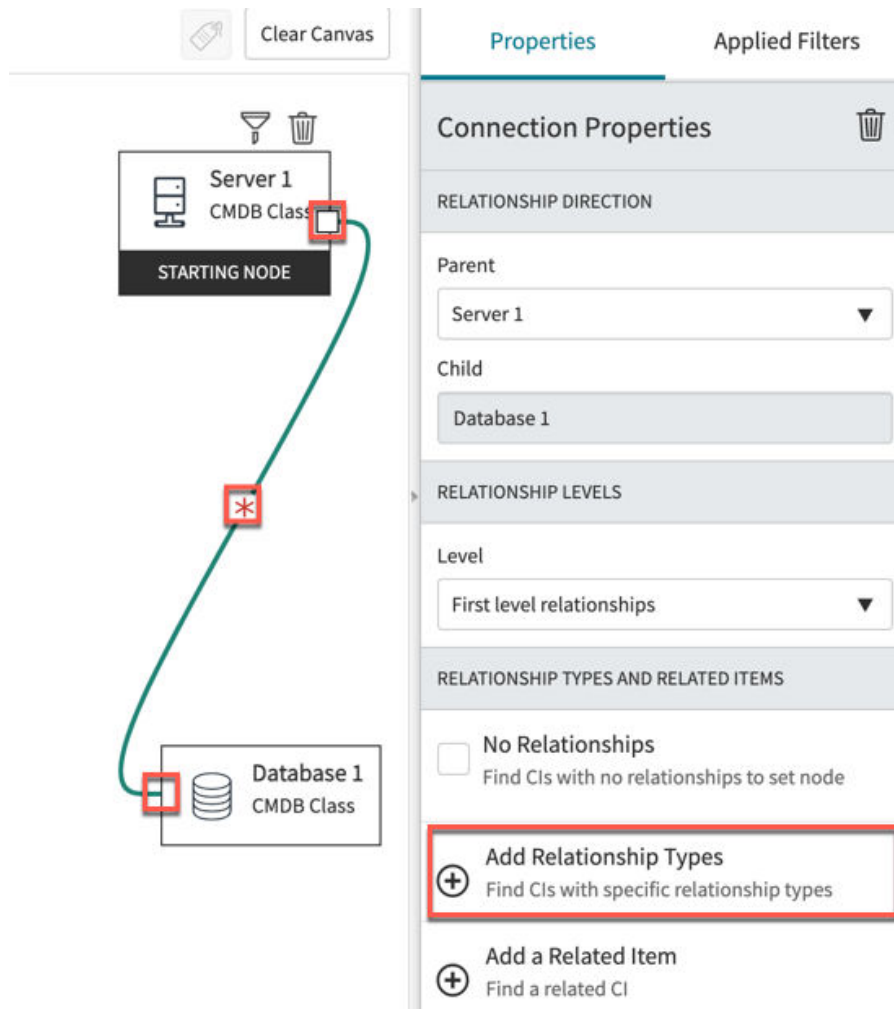
Role required: none

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. Select **Create new** and then enter a name, for example **All servers with a connection to a DB**.
3. Choose **CMDB Query** and then select **Create**.
4. In the **CMDB Classes** list, locate the **Server** class and drag it to the canvas.

Tip: Use the search box to find items quickly.

5. Locate the **Database** class, and place it to the right of the **Server** class node on the canvas.
6. Select at the center of the right side of **Server**, and then at the center of the left side of **Database** to create a connection line between the two class nodes.
7. Select once or twice on the connection line until the Connection Properties panel appears in the sidebar.
8. In the Relationship Types and Related Items section, select **Add Relationship Types** and add all the relationships from the list.



Settings in the Relationship Direction section reflect the parent-child direction in the relationship. If the Database class is the parent in the relationship, then the **Parent** and **Child** settings are switched.

9. Select **Save**, and then select **Saved Queries** on the left to see the widget for the saved query.
10. Select the query widget to return to the canvas in edit mode.
11. Select **Run** to execute the query.

Review the query results. Each row displays the name of a server CI, the name of a database CI, and the relationship type between them.

12. Add columns to the query results.
 - a. Select the **Server 1** node on the canvas once or twice so that the Server 1 Report Columns section appears in the right-side pane.
 - b. Select **Add Columns**.
 - c. Select **Manufacturer** and then select outside the columns list to close it.
 - d. Select **Run**.
The query results now include the **Manufacturer** column.
 - e. Select **Save** again to save your customization for this query.

CMDB — Critical application services

Use this example to build a CMDB query to find all critical application services, and their owner.

Before you begin

Role required: none

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder** and then select **Create new**.
2. Enter All critical application services as the query **Name**.
3. Choose **CMDB Query** and then select **Create**.
4. In the **CMDB Classes** list, locate the **Application Service** class and then drag it to the canvas.

Tip: Use the search box to find items quickly.

5. Add a filter to the application service node.

- a. Point to the application service node and then select the **Apply filters** icon.
 - b. In the Filters section, add the condition **[business criticality] [is] [1 - most critical]**.
 - c. Close the Filters section.
6. Add columns to the query results.
 - a. In the Properties sidebar, select **Add Columns**.
 - b. Select **business criticality** and **owned by**, and then select outside the columns list to close it.
 7. Select **Save**.
 8. Select **Run** and then review the results.
You can, for example, locate any of the critical application services without an owner.

CMDB — Application services with incident or change

Use this example to build a CMDB query to find all application services for which there is an incident or a change request. Results are returned for either the application service itself or any CI within the service.

Before you begin

Role required: none

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder**.
2. Select **Create new** and then enter **Application services with incidents or change requests** as the query name.
3. Choose **CMDB Query** and then select **Create**.
4. In the **CMDB Classes** list, locate the **Application Service** class and then drag it to the canvas.

Tip: Use the search box to find items quickly.

5. Select **Non-CMDB Tables**.
6. Locate the **Incidents** class in the class hierarchy, and then drag it to the canvas.
7. Locate the **Change Requests** class in the class hierarchy, and then drag it to the canvas.
8. Connect the Application Service and the Incidents nodes, and then, in the Properties right-side bar.
 - a. Select **Apply Incidents reference filter to all nodes in the pattern**.
 - b. Set **Use CI reference column** to **Configuration item**.
9. Connect the Application Service and the Change Request node, and then, in the Properties right-side bar.
 - a. Select **Apply Change Request reference filter to all nodes in the pattern**.
 - b. Set **Use CI reference column** to **Configuration item**.
10. Select the **And** operator between the Incidents and the Change Request nodes, and switch it to **Or**.
11. Select **Save**.
12. Select **Run** and then review the results.

CMDB — Hardware with Windows installed

Use this example to build a CMDB query to find all hardware in my service offering that has Microsoft Windows installed.

Before you begin

Role required: none

Procedure

1. Navigate to **All > Configuration > CMDB Query Builder** and then select **Create new**.

2. Enter `All hardware in my service offering that has Windows installed` as the query name.
3. Choose **CMDDB Query** and then select **Create**.
4. In the **CMDDB Classes** list, locate the following classes, and then drag them to the canvas:

Tip: Use the search box to find items quickly.

- **Service**
- **Service Offering**
- **Application Service**
- Searching for infrastructure, **Hardware**

5. Connect the Service node to the Service Offering node.

In the Properties sidebar, select **Add Relationship Type** and select the **Connect to::Connected by** relationship.

6. Connect the Service Offering node to the Application Service node.

In the Properties right-side bar, select **Add Relationship Type** and select the **Connect to::Connected by** relationship.

7. Select the Application Service node.

In the Properties right-side bar, select **Convert attached nodes to pattern** to include all CIs within the application service, in the query.

8. Connect the Application Service node to the Hardware node.

9. All infrastructure under Service,

10. Select **Save**.

11. Select **Run** and then review the results.

You can select **Column options** of the Service column header, and select to **Group by Service**. Then expand a service to see all the hardware infrastructure under that service.

12. Return to the CMDB Query Builder window, to expand the query to include only infrastructure CIs on which Windows is installed.
13. Select **Non-CMDB Tables**, locate the **Software Instance** class, and drag it to the canvas.
14. Connect the Hardware node to the Software Instance node.

In the Properties right-side bar, set **Use CI reference column** to **Installed on**.
15. Point to the Software Instance node, and select on the **Apply filters** icon that appears.
 - a. In the Filters section, add the condition **[Product Name.Name] [is] [windows]**.
 - b. Close the Filters section.
16. Select **Save**.
17. Select **Run** and review the new results.

Service Mapping — Linux servers in services

Use this example to build a Service Mapping query to find all Linux servers in services.

Before you begin

Role required: none

About this task

This example Service Mapping query finds all services that include Linux servers. Then, with a simple modification, the query can find all services that do not include a Linux server.

Procedure

1. Navigate to **All > Configuration** and select **CMDB Query Builder**.
2. Select **Create new** and then enter `Linux server in services` in the **Name** field.

3. Choose **Service Mapping Query** and then select **Create**.
4. In the **CMDB Classes** hierarchy list, locate **Linux Server** and drag it to the canvas.

Tip: Use the search box to find items quickly.

5. Select **Run**.

Review the query results. Each row displays the name of a Service Mapping service and the name of a Linux server that is a member of that service.

6. On the right-side pane, disable **Services Including This Pattern** and then select **Run** again.

Review the query results. Now, each row displays the name of a Service Mapping service that doesn't include the specified Linux server.